

BUSINESS ANALYTICS – WEEK 7 TASK

Unsupervised Financial Anomaly Detection via Isolation Forest from First Principles

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Task Objective

The objective of this task is to detect unusual or suspicious financial transactions using the **Isolation Forest algorithm**. The algorithm identifies transactions that are significantly different from normal transaction patterns by isolating them through random data partitions.

1. Problem Statement

Financial systems process a large number of transactions every day. Most transactions follow normal patterns, while a small number may be unusual because of high transaction amounts, unusual transaction frequency, unexpected transaction times, or abnormal account activity.

The objective is to develop an **unsupervised anomaly detection approach** that can identify unusual transactions without requiring previously labelled fraud data. Isolation Forest works by randomly selecting transaction attributes and creating binary partitions. Transactions that are isolated using fewer partitions are considered more likely to be anomalies.

2. Input Data

Field	Example	Description
Transaction_ID	T10025	Unique transaction identifier
Account_ID	A501	Unique account identifier
Transaction_Amount	85000	Amount of transaction
Transaction_Frequency	15	Number of transactions
Account_Balance	120000	Current account balance
Transaction_Hour	2	Hour of transaction
Transaction_Type	Transfer	Type of transaction

The numerical attributes are used to identify unusual transaction behaviour.

3. Isolation Forest Methodology

Step 1 – Data Ingestion: Collect the financial transaction records from the available database or dataset.

Step 2 – Data Cleaning: Remove duplicate records, handle missing values, correct invalid values, and prepare the transaction attributes for analysis.

Step 3 – Feature Selection: Select relevant attributes such as transaction amount, transaction frequency, account balance, and transaction time.

Step 4 – Random Sampling: A random subset of transactions is selected for constructing each Isolation Tree.

Step 5 – Random Feature and Split Selection: A feature is randomly selected and a random split value is chosen. For example: **Transaction Amount < 20,000**. The transactions are divided into two groups based on this condition.

Step 6 – Recursive Binary Partitioning: The two groups are repeatedly divided using new randomly selected features and split values. This process continues until transactions are isolated or the required tree depth is reached.

Step 7 – Construct Isolation Trees: Several Isolation Trees are created using different random samples and random partitions.

Step 8 – Construct Isolation Forest: The collection of Isolation Trees forms the **Isolation Forest**. The results from all trees are combined to calculate the average path length for each transaction.

4. Path Length and Anomaly Detection

Path length represents the number of splits required to isolate a transaction. For example, if a transaction is isolated after three splits, the path length is 3.

Short path → **More likely anomaly**

Long path → **More likely normal**

5. Mathematical Anomaly Score

The Isolation Forest anomaly score is calculated using:

$$s(x,n) = 2^{-E(h(x)) / c(n)}$$

Where: **s(x,n)** = anomaly score; **E(h(x))** = average path length of transaction x; **c(n)** = average path length for a sample of size n; **n** = number of observations.

Score	Interpretation
Close to 1	More likely to be anomalous
Around 0.5	Boundary / uncertain
Lower score	More likely to be normal

6. Small Worked Example

Transaction	Amount	Frequency	Classification
T1	■1,000	2	Normal
T2	■1,500	3	Normal
T3	■2,000	2	Normal
T4	■2,500	4	Normal
T5	■2,00,000	30	Potential Anomaly

T1–T4 have similar transaction behaviour, while T5 is significantly different. During random partitioning, T5 may be separated from the other transactions in a small number of splits. Therefore, T5 will have a shorter average path length and a relatively higher anomaly score. This makes T5 a candidate for further investigation.

7. Suggested Implementation Workflow

Financial Transaction Data → Data Cleaning → Feature Selection → Random Sampling → Random Feature & Split → Recursive Partitioning → Isolation Trees → Isolation Forest → Average Path Length → Anomaly Score → Normal / Anomalous Transactions → Business Analysis

8. Expected Business Insights

- Identify unusual financial transactions.
- Detect abnormal spending patterns.
- Identify accounts with unusual transaction behaviour.
- Reduce the amount of manual transaction checking.
- Support financial risk monitoring.
- Discover previously unknown transaction patterns.

An anomaly detected by the algorithm should be treated as a **potentially unusual transaction**, not automatically as confirmed fraud. Further investigation may be required.

9. Tools / Technologies

Recommended tools include: **Python** for algorithm implementation and analysis, **Pandas** for data processing, **NumPy** for mathematical calculations, **SQL** for transaction data management, **Apache Spark / PySpark** for large-scale data processing, and **Power BI / Tableau** for visualization.

10. Deliverables

- Cleaned financial transaction dataset
- Selected transaction features
- Isolation Tree structure
- Isolation Forest
- Path length calculations
- Average path length
- Anomaly scores
- Normal and anomalous transaction results
- Anomaly visualization
- Business insights
- Final conclusion

11. Conclusion

Isolation Forest provides an unsupervised method for identifying unusual financial transactions. It works by randomly partitioning transaction data and measuring how quickly individual transactions can be isolated.

Transactions that require fewer partitions generally have shorter path lengths and higher anomaly scores. By combining multiple Isolation Trees into an Isolation Forest, the method provides a systematic approach to detecting unusual financial behaviour.

This approach can support financial monitoring, risk analysis, and further investigation of potentially abnormal transactions.